

Ticketing System Lab – Peppermint

Author:

Sivateja Mattaparthi

Role Focus

IT Support • Help Desk • Service Desk • Desktop Support

Lab Objectives

- Gain hands-on experience with an enterprise help desk ticketing system.
- Create, assign, prioritise, update, and resolve support tickets.
- Understand the complete incident management lifecycle.
- Document troubleshooting activities and technician notes.
- Manage ticket status and communication throughout the support process.
- Develop practical IT Service Management (ITSM) skills used in Help Desk environments.

Component	Details
Host Operating System	macOS Catalina (10.15.8)
Platform	MacBook Air (2019 Retina)
Container Platform	Docker Desktop
Container Runtime	Docker Compose
Ticketing System	Peppermint
Database	PostgreSQL 16
Deployment	Localhost (Port 3000)
Environment	Home Lab
Tools Used	Docker Desktop, Docker Compose, Peppermint, PostgreSQL, Web Browser

Introduction

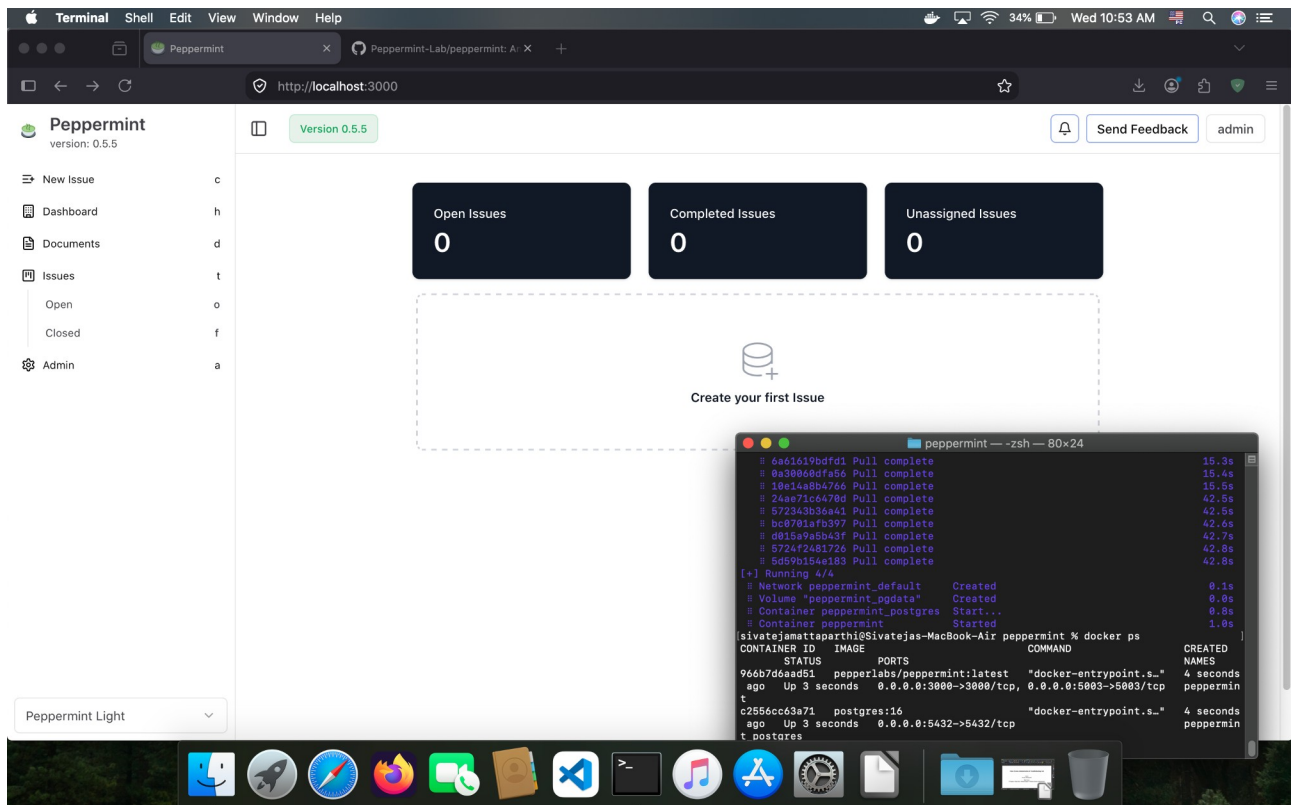
Modern IT Support teams rely on ticketing systems to manage, prioritise, and resolve user requests efficiently. This hands-on lab demonstrates the deployment of the Peppermint help desk platform using Docker and PostgreSQL, followed by practical experience creating, assigning, updating, and resolving support tickets. The exercises are to simulate common IT Service Management (ITSM) workflows performed by Help Desk and Service Desk professionals in enterprise environments.

Peppermint Deployment & Initial Setup

Peppermint was deployed locally using Docker Desktop with a PostgreSQL 16 database container. After the containers were successfully started, the application was accessed through localhost (port 3000/3003), where the initial dashboard confirmed that the ticketing system was running correctly and ready for configuration.

Components Verified

- Docker containers running successfully.
- Peppermint web application accessible via localhost.
- PostgreSQL database connected and operational.
- Initial dashboard loaded with no active issues.
- Environment ready for ticket creation and ITSM workflow testing.



I created internal users and clients within the Peppermint Help Desk platform to simulate a real-world IT support environment as the system administrator.

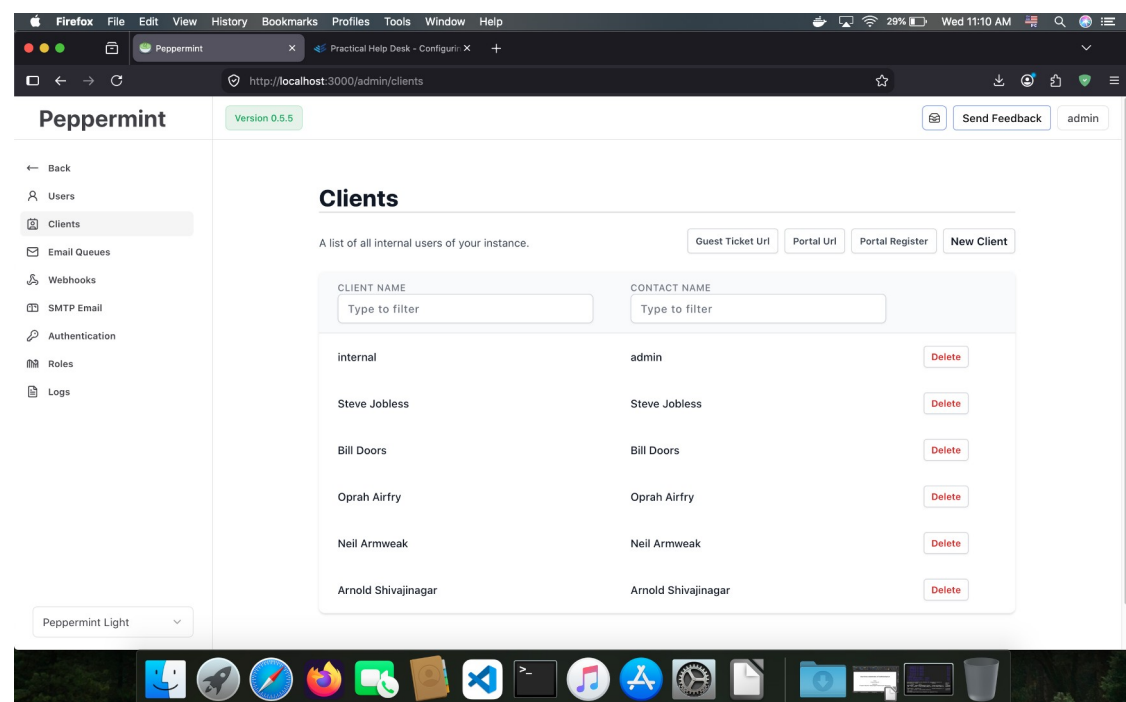
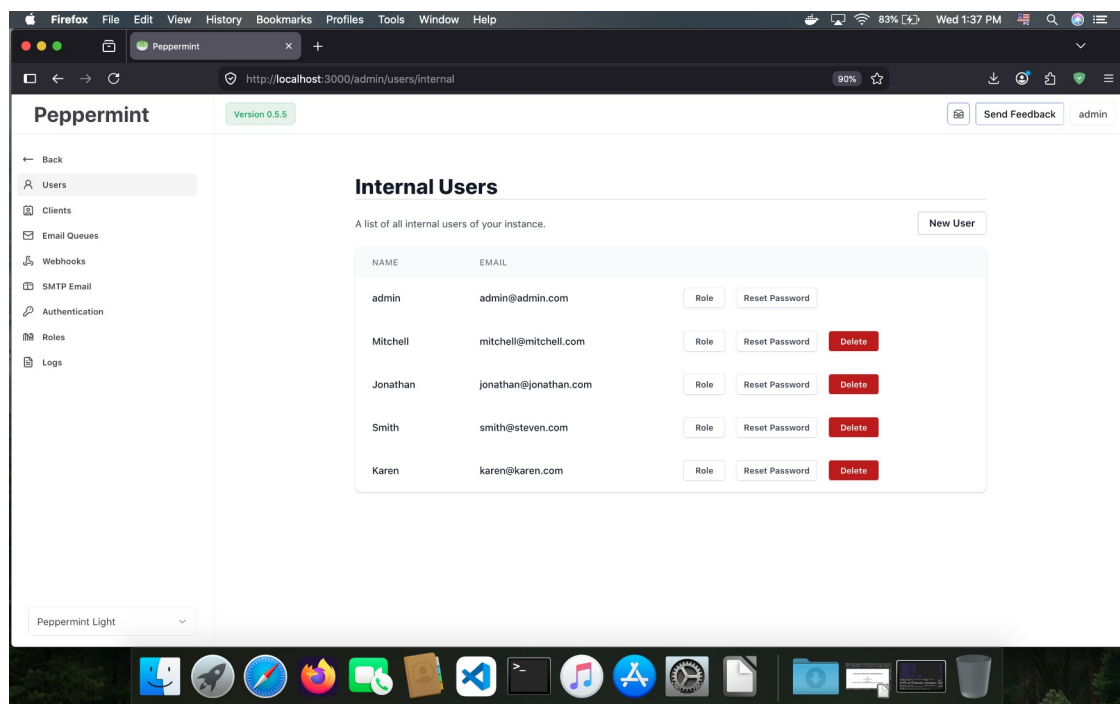
Internal Users

Internal users represent the IT support staff who manage and resolve service requests within the help desk system. They are responsible for creating, assigning, updating, and closing support tickets, as well as communicating with end users throughout the troubleshooting process.

Administrators can also manage user accounts, reset passwords and configure roles and permissions to control access to the platform.

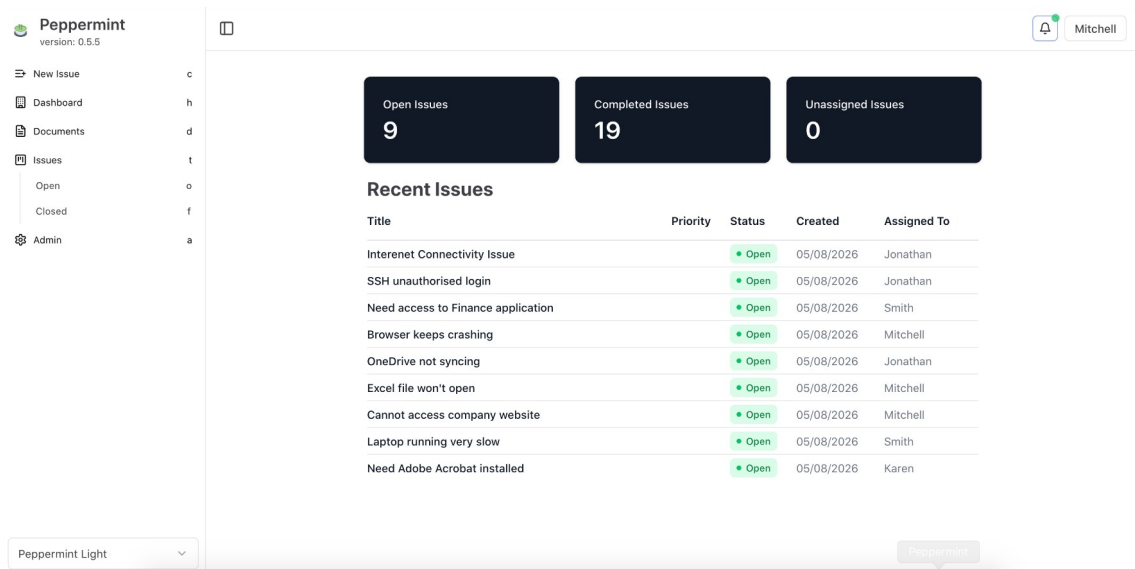
Clients

Clients represent the end users or employees who request technical assistance from the IT help desk. They create support tickets for issues such as hardware, software, network, or account-related problems, allowing the support team to track, prioritise and resolve requests efficiently. Maintaining client records helps organise support requests and ensures clear communication throughout the ticket lifecycle.



Help Desk Dashboard Overview

Once users begin submitting support requests, the dashboard provides a real-time overview of the help desk's workload. It displays the number of open, completed, and unassigned tickets, allowing technicians to quickly assess the current support queue. Recent tickets show their status, creation date, and assigned technician, enabling the team to monitor unresolved issues, track completed requests, and manage ticket assignments efficiently.

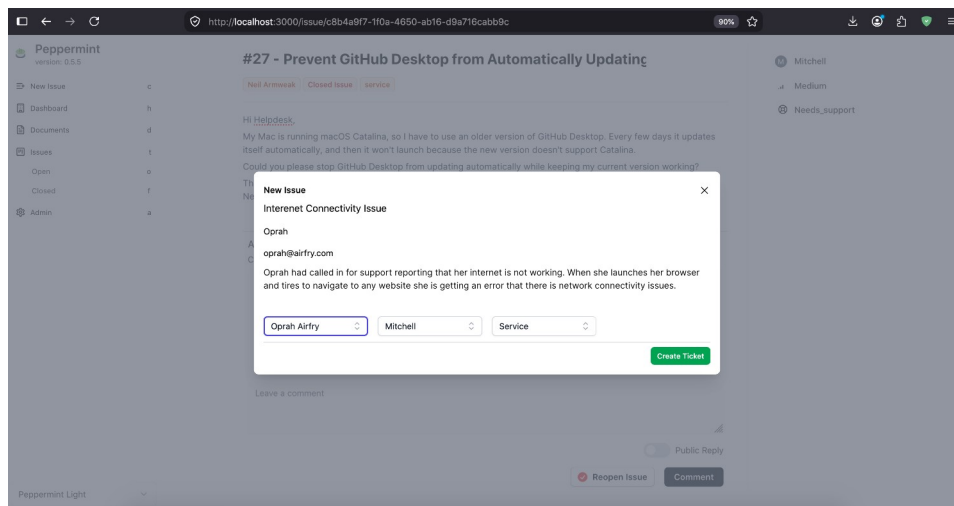


Creating a Support Ticket

To simulate a real-world IT Help Desk workflow, a new support ticket was created within the Peppermint Help Desk platform. Each field captures important information required for efficient ticket management and assignment.

Fields Used

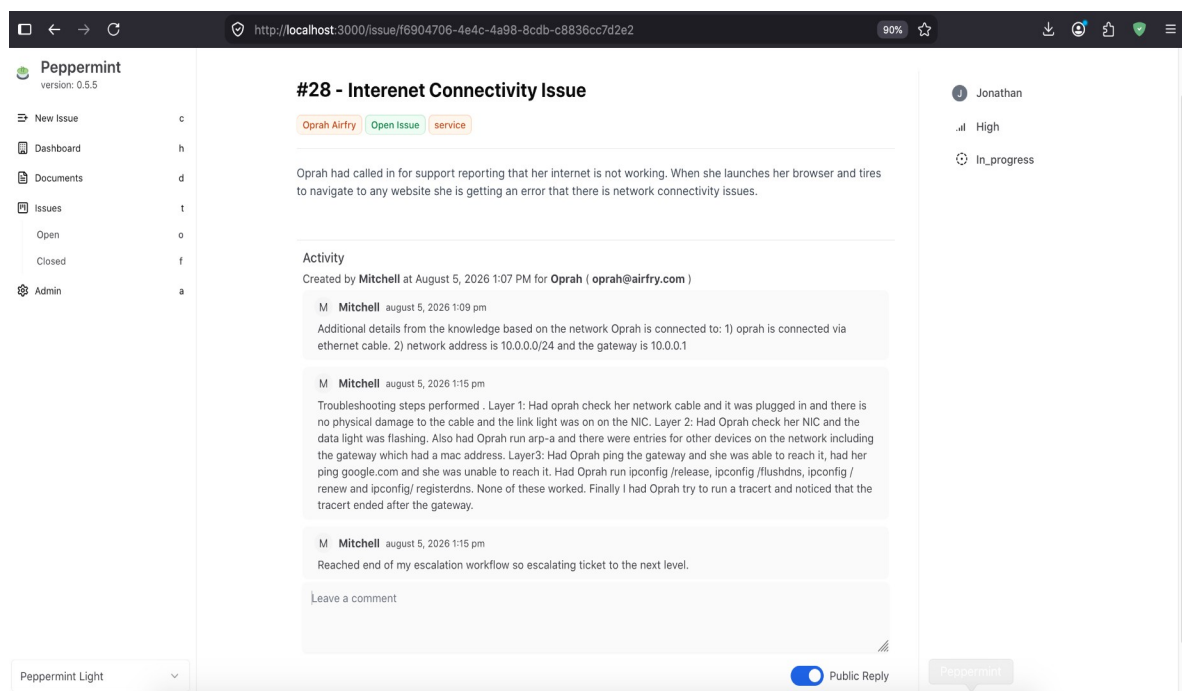
- **Ticket Title** – A brief summary of the issue being reported (e.g., *Internet Connectivity Issue*).
- **Client Name** – The end user experiencing the issue and requesting IT support.
- **Client Email** – Contact email used for communication and ticket updates.
- **Issue Description** – A detailed explanation of the problem, including symptoms and any relevant information to assist the technician.
- **Assigned To** – The IT Help Desk technician responsible for investigating and resolving the ticket.
- **Category / Service** – Classifies the ticket based on the type of request (e.g., Service, Incident, Hardware, Software), helping with organization and reporting.
- **Create Ticket** – Submits the request and adds it to the help desk queue for tracking and resolution.



Ticket Investigation and Troubleshooting

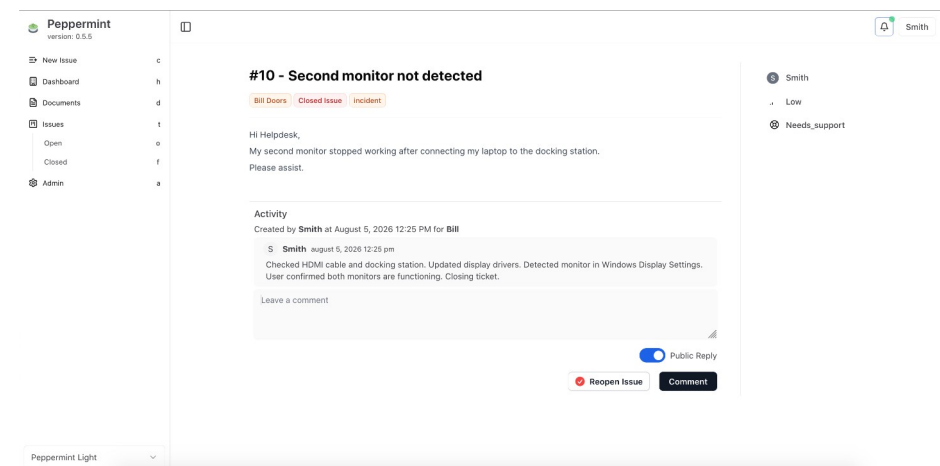
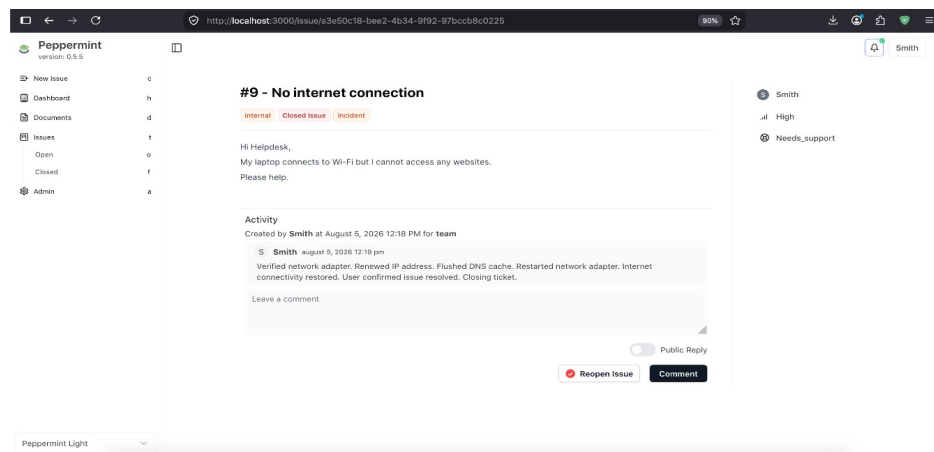
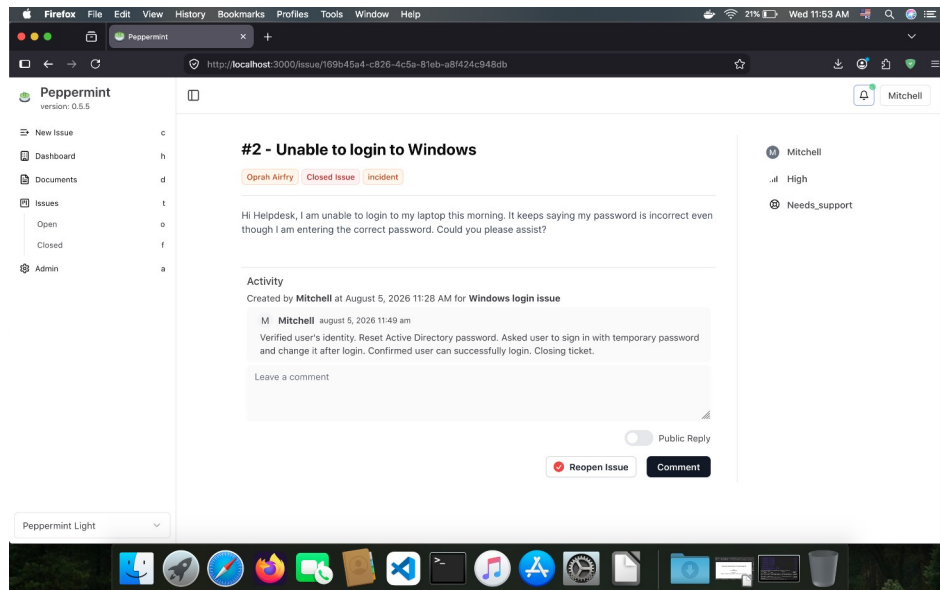
After receiving the support ticket, the issue was investigated using a structured Layer 1 to Layer 3 troubleshooting approach. Layer 1 (Physical Layer) verification included checking the Ethernet cable connection, NIC link lights, and ensuring there was no physical damage. Layer 2 (Data Link Layer) focused on verifying the network adapter status and examining the ARP table to confirm communication with devices on the local network. Layer 3 (Network Layer) involved validating the IP configuration, default gateway, and testing connectivity using tools such as `ping`, `ipconfig`, and `tracert` to isolate the point of failure. All troubleshooting steps and observations were documented within the ticket to maintain a complete support history.

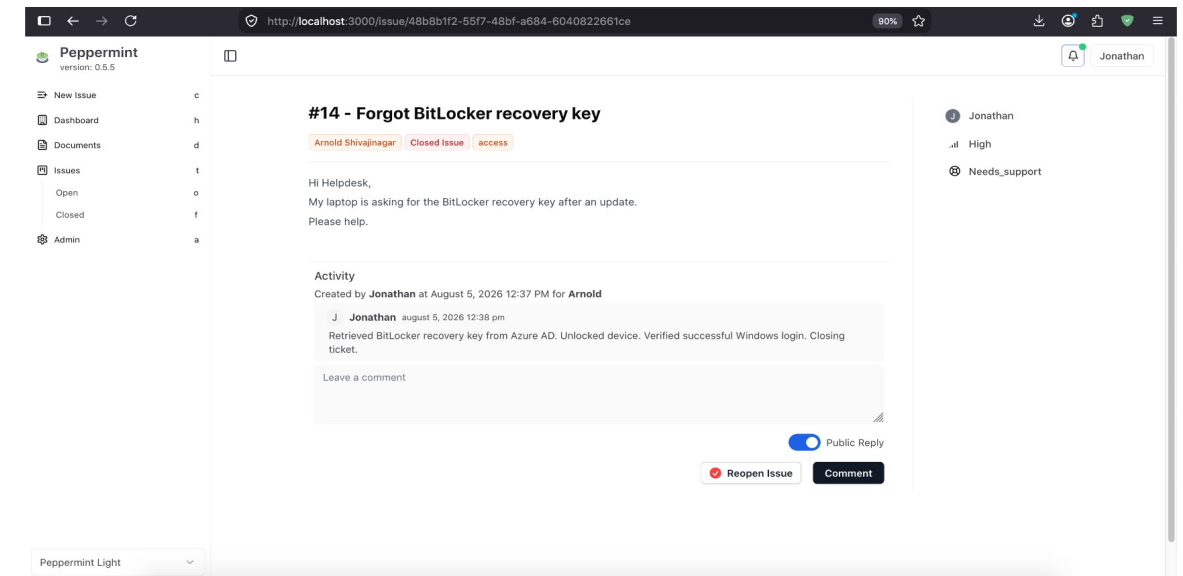
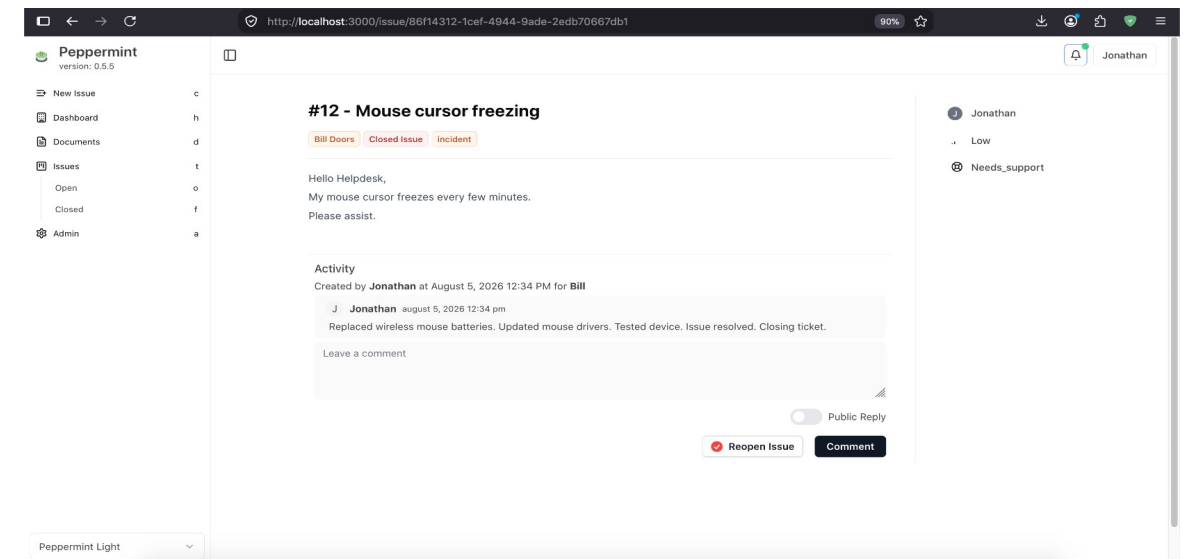
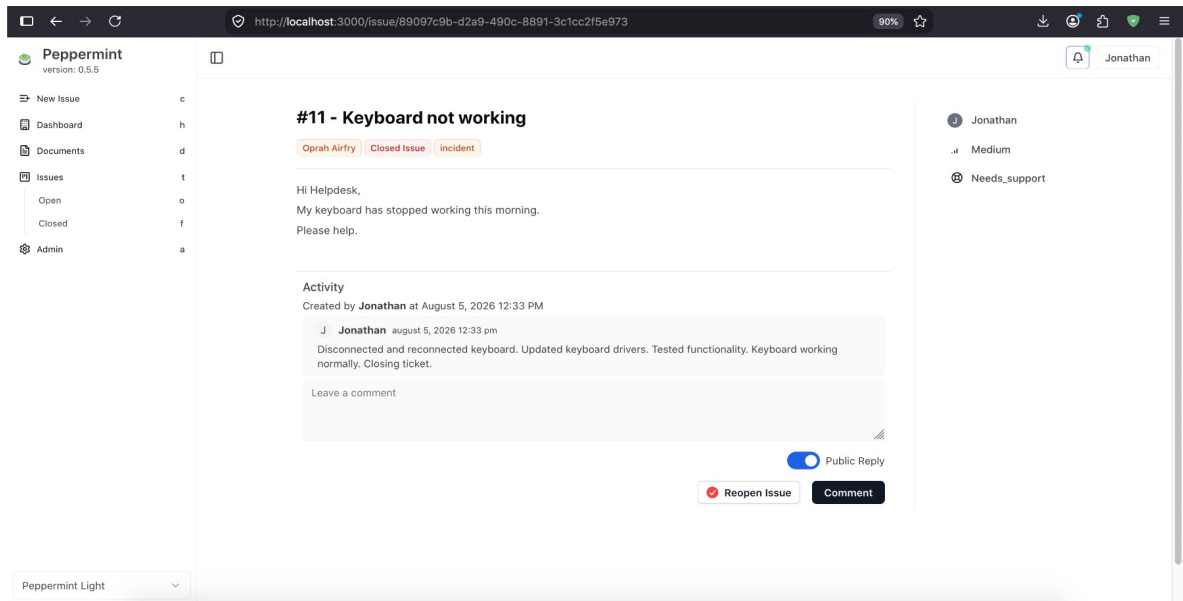
As the issue could not be resolved after completing all standard Level 1 troubleshooting procedures, the incident was escalated to the next support tier with a High priority. The ticket activity log provides a complete record of the investigation, troubleshooting performed, and escalation, ensuring a seamless handover for further analysis by senior support engineers.



Sample Resolved Service Desk Tickets

The following examples demonstrate a selection of service desk tickets that were investigated, documented and successfully resolved using standard IT Help Desk troubleshooting procedures.





Peppermint
version: 0.5.5

New Issue

Dashboard

Documents

Issues

Open

Closed

Admin

Jonathan

#15 - Need access to Shared Drive

Oprah AirfryClosed Issueaccess

Hi Helpdesk,
I need access to the Finance shared drive for my new project.
Thanks.

Activity

Created by Jonathan at August 5, 2026 12:39 PM for Oprah

JJonathan

august 5, 2026 12:39 pm

Verified manager approval. Added user to Finance security group. Confirmed access to shared drive. Closing ticket.

Leave a comment

Public Reply

Reopen IssueComment

Peppermint Light

Peppermint
version: 0.5.5

New Issue

Dashboard

Documents

Issues

Open

Closed

Admin

Karen

#17 - Zoom camera not working

Steve JoblessClosed Issuebug

Hi Helpdesk,
My camera isn't working during Zoom meetings.
Please help.

Heading

Activity

Created by Karen at August 5, 2026 12:42 PM for Steve

KKaren

august 5, 2026 12:43 pm

Enabled camera permissions. Updated webcam drivers. Performed Zoom test meeting. Camera functioning correctly. Closing ticket.

Leave a comment

Public Reply

Reopen IssueComment

Peppermint Light

Peppermint
version: 0.5.5

New Issue

Dashboard

Documents

Issues

Open

Closed

Admin

Smith

#23 - Need access to Finance application

Arnold ShivajinagarClosed Issuebug

Hi Helpdesk,
I recently joined the Finance department and need access to the Finance application.
Thank you.

Activity

Created by Smith at August 5, 2026 12:53 PM for Arnold

SSmith

august 5, 2026 12:53 pm

Verified manager approval. Assigned required application role. Confirmed successful login. Closing ticket.

SSmith

august 5, 2026 1:35 pm

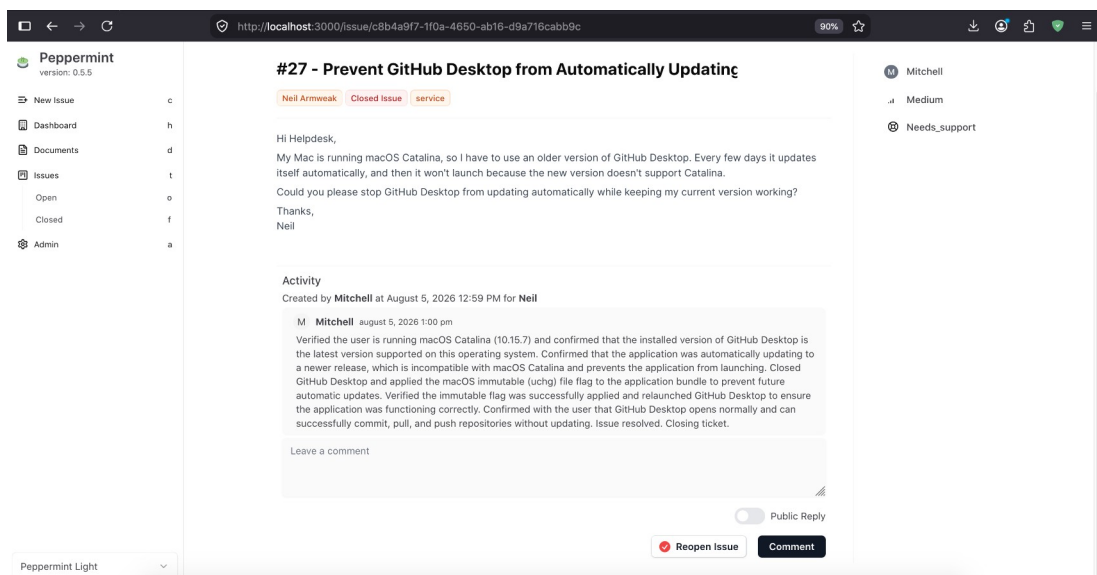
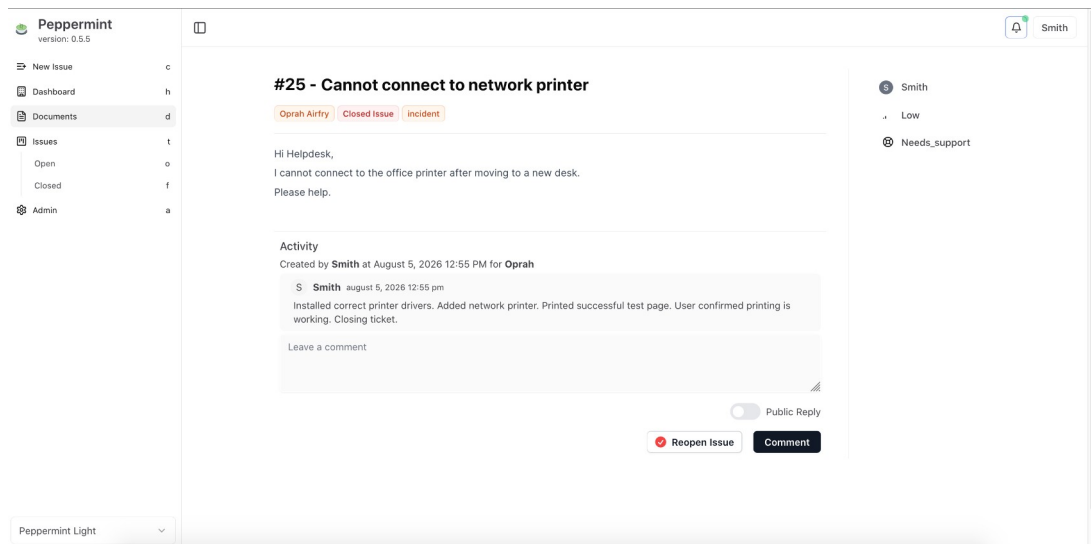
Verified manager approval. Assigned required application role. Confirmed successful login. Closing ticket.

Leave a comment

Public Reply

Reopen IssueComment

Peppermint Light



Note:

A small number of ticket screenshots are unavailable because the Docker containers and volumes were accidentally removed during the lab, which resulted in the loss of the local Peppermint Help Desk instance and its data. The remaining screenshots accurately demonstrate the ticketing workflows, troubleshooting procedures, and resolution process completed during the lab.

Escalated Tickets Section:

The following examples demonstrate support tickets that were initially investigated and troubleshooted by the Level 1 Help Desk. After completing all standard troubleshooting procedures and documenting the findings, the issues could not be resolved at the first support level. As a result, the tickets were escalated to higher-level support teams for advanced investigation and resolution. This reflects a common IT service desk workflow, where unresolved incidents are escalated with complete troubleshooting notes to ensure an efficient handover and continuity of support.

Peppermint
version: 0.5.5

New Issue
Dashboard
Documents
Issues
Open
Closed
Admin

#6 - Unable to connect to VPN

Arnold Shivajinagar | Closed Issue | service

Hello Helpdesk,
I am working from home today and I am unable to connect to the company VPN.
The VPN client displays an authentication error each time I try to connect.
I have already restarted my laptop but the issue remains.
Please assist.

Activity
Created by **Mitchell** at August 5, 2026 11:57 AM for **Arnold**

M Mitchell august 5, 2026 11:58 am
Verified user's account is active. Confirmed internet connectivity is working. Reset VPN profile. Reinstalled VPN client. Retested connection. Authentication error continues. Unable to resolve at Level 1. Escalating to Network Team for further investigation.

J Jonathan august 5, 2026 12:02 pm
Received escalated ticket from Level 1 Helpdesk. Reviewed VPN gateway logs and found the user's account was failing due to an expired MFA token. Reset the user's VPN profile on the VPN gateway. Cleared cached credentials. Asked the user to sign out and sign back into the VPN client. Verified successful VPN connection. User confirmed access to internal resources. Issue resolved. Received escalated ticket from Level 1 Helpdesk. Reviewed VPN gateway logs and found the user's account was failing due to an expired MFA token. Reset the user's VPN profile on the VPN gateway. Cleared cached credentials. Asked the user to sign out and sign back into the VPN client. Verified successful VPN connection. User confirmed access to internal resources. Issue resolved. Closing ticket.

Leave a comment

Peppermint Light

Peppermint
version: 0.5.5

New Issue
Dashboard
Documents
Issues
Open
Closed
Admin

#8 - Need Adobe Acrobat installed

Neil Armweak | Closed Issue | service

Hi Helpdesk,
Could you please install Adobe Acrobat Pro on my work laptop? I need it to edit PDF documents.

Activity
Created by **Smith** at August 5, 2026 12:09 PM for **Steve Jobless**

S Smith august 5, 2026 12:09 pm
Verified software request. Attempted software installation. Installation failed because the required enterprise license is unavailable. Unable to complete request at Level 1. Escalating to Software Packaging Team.

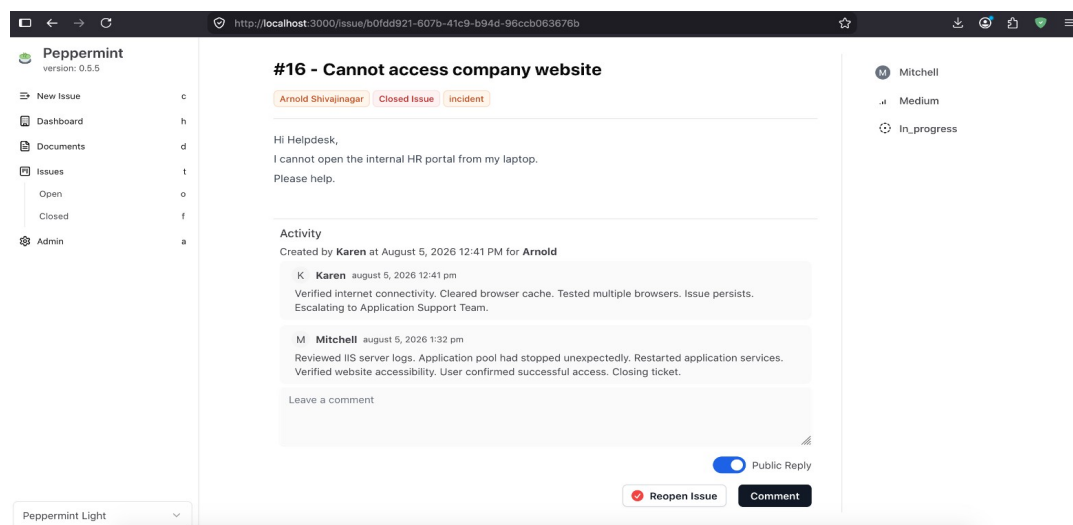
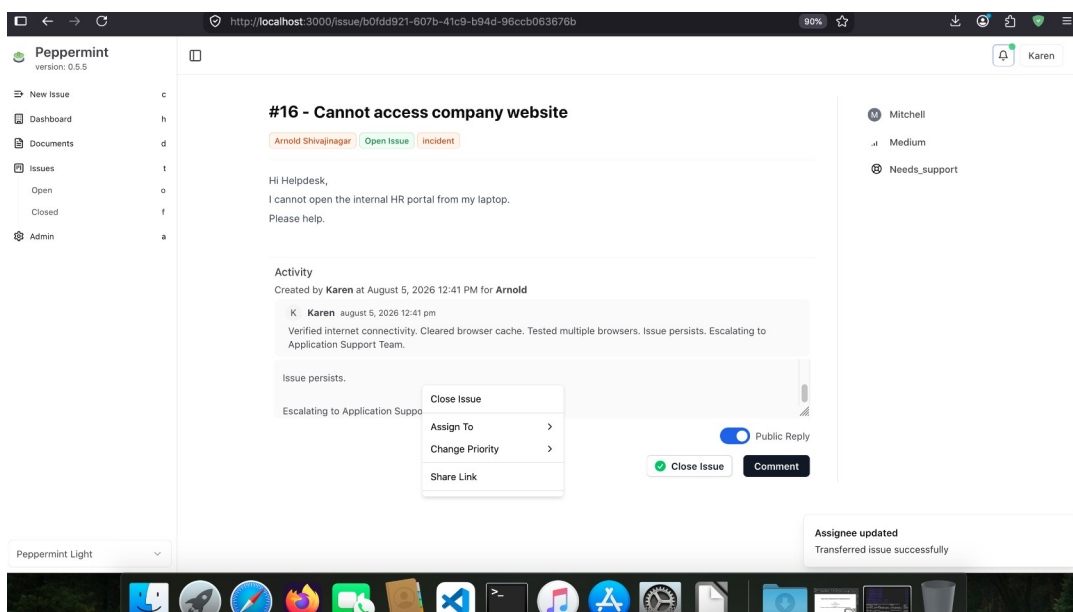
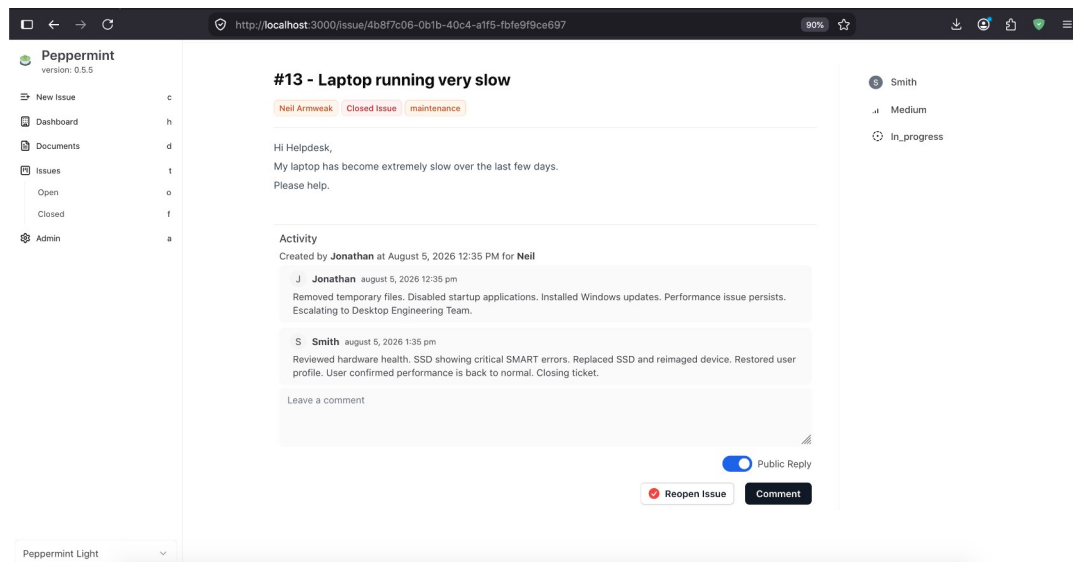
K Karen august 5, 2026 1:36 pm
Reviewed hardware diagnostics. Memory module failed testing. Replaced faulty RAM. Performed stress testing for 45 minutes. No further blue screen errors observed. User confirmed laptop is working normally. Closing ticket.

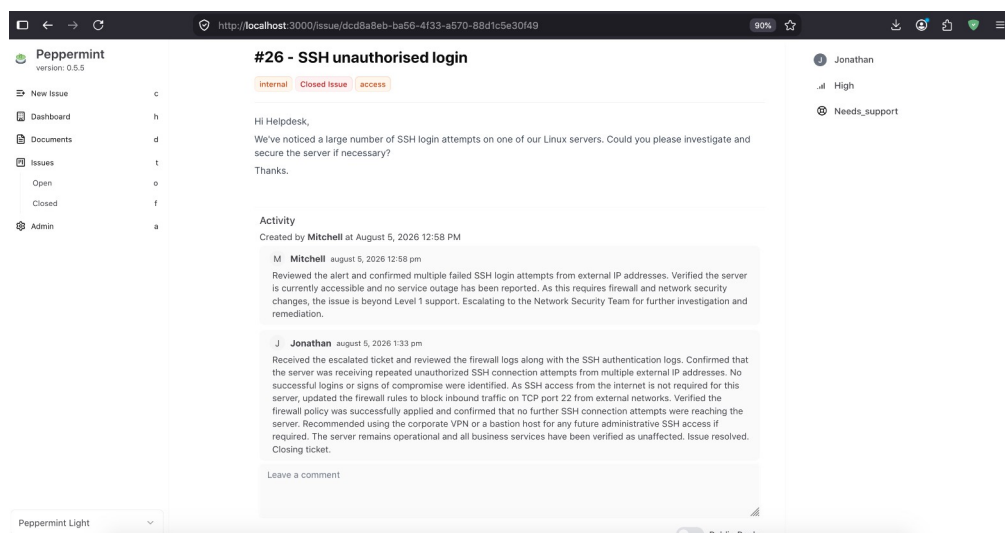
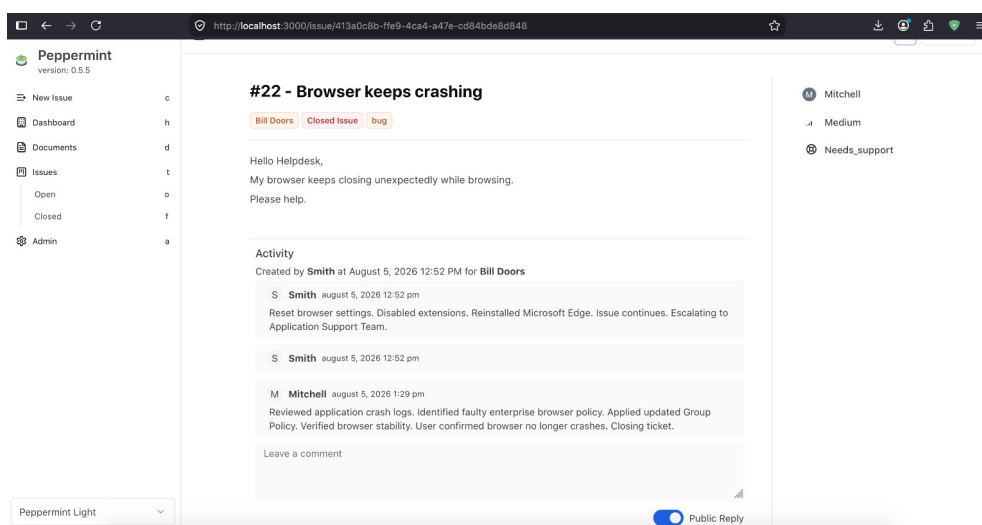
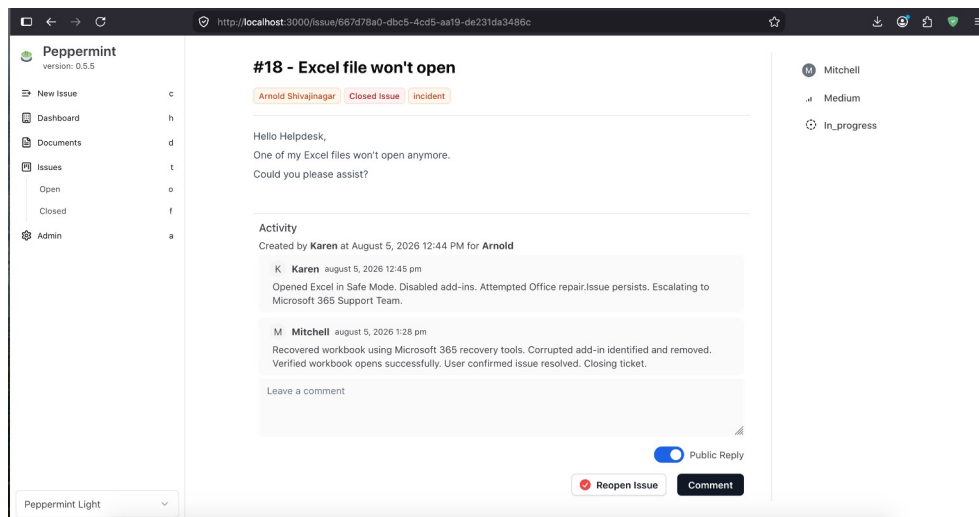
Leave a comment

Public Reply

Reopen Issue | Comment

Peppermint Light





Conclusion: This lab demonstrates standard IT Help Desk ticketing workflows. In a real enterprise environment, support requests are typically more complex and involve additional systems, processes, and collaboration across multiple support teams. More advanced ticketing scenarios and enterprise-focused labs will be added to this portfolio as they are completed.